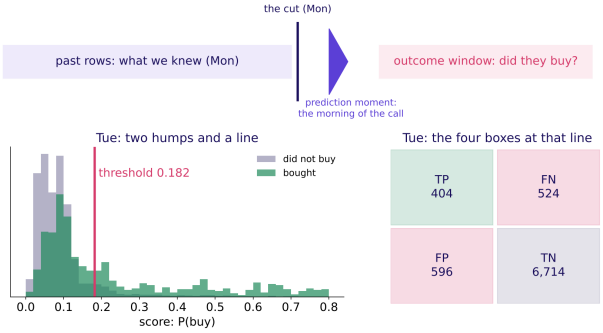
The metric trap: cheat sheet

Week 5, Tuesday. Two humps, one line, four boxes, two prices.



Which terms carry the day?

Term	Plain meaning	Rule or formula	Trap
Confusion matrix	counts of (what the model said, what happened)	TP, FP, FN, TN	scikit-learn prints `[[TN, FP], [FN, TP]]`
Precision	of those flagged, the share right	`TP / (TP + FP)`	dividing by everyone
Recall	of those real, the share flagged	`TP / (TP + FN)`	high precision hiding a recall of 0.19
F1	one grade, low if either is low	`2PR / (P + R)`	weighs both mistakes equally
Accuracy on imbalance	share right, mostly the easy majority	say-no scores `1 - base rate`	88.74 percent for calling nobody
1-in-N test	act when at least 1 in N would come good	`wrong action / (wrong action + missed action)`, each in Rs	needs honest probabilities
Threshold	the score at or above which we act	from the prices, or from the budget if stricter	0.5 is the line for equal costs only
Precision at k	share of positives in the top k	`positives in top k / k`	comparing lists of different k

Where is today on the machine learning map?



Which real numbers make it stick?

- Stripe: on a 26 dollar sale at 8 percent margin, one fraud costs the profit of 18.71 good sales, so blocking pays once about 1 in 20 blocks is fraud, a break-even precision of 5.07 percent (Radar technical guide, verified 28 October 2026).

- Flipkart with CMU: of the 250 accounts a ratings-fraud model ranked most suspicious, Flipkart's team found 211 to be fraud, a precision at 250 of over 84 percent (arXiv 1511.06030, verified 28 October 2026).
- Razorpay: fraud detection is compared "at the same alert volume", since catching more by reviewing more "is a threshold change, not proof that the model improved" (engineering blog, 10 September 2026, verified 28 October 2026).

What has the programme covered so far, one line each?

Day	One line
W1 Mon	The revenue tree: a business ask split into measurable branches with honest first numbers in Python.
W1 Tue	The sales-drop ladder: like-with-like windows, denominators and descriptive statistics by segment.
W1 Wed	Profile, clean and reconcile before any analysis, so input equals clean plus rejected.
W1 Thu	Real or noise with a permutation test, correlation against cause, and the one-page note.
W1 Fri	The week rebuilt alone, with no AI, and the note defended aloud.
W2 Mon	SQL from selection to CTEs: the revenue tree as warehouse queries.
W2 Tue	Joins without lying: fan-out, the anti-join, booked against collected.
W2 Wed	Window functions: ranks, running totals and spend against plan.
W2 Thu	pandas groupby, merge and reshape for the Monday customer table.
W2 Fri	Excel for analysts: pivots, lookups, one number on the leadership slide.
W3	Build 1: a named-customer brief on raw messy data, defended live.
W4 Mon	Metric design: a north star with guardrails, and the proxy test.
W4 Tue	Stakeholder communication: one slide, five minutes, and the first guesstimate.
W4 Wed	Market basket: support, confidence, lift, and cross-sell value in rupees.
W4 Thu	Cohort and funnel: where the frequency loss actually happens.
W4 Fri	Forecasting against a naive baseline that is hard to beat.
W5 Mon	Framing a prediction: input, target, prediction time, the split, the baseline, two model families.
W5 Tue	The metric trap: the matrix, the ratios, accuracy on imbalance, prices to a line, the transfer to lending.

When does each rule apply, and what flips it?

Situation	Report or do	Flips when
Miss dearer than a false action (Rs 1,200 against Rs 40)	recall leads, line low: 0.032	the ratio tightens: at 8 to 1 the line is 0.11
False action dearer (Rs 60,000 against Rs 3,000)	precision leads, line high: P(repay) 0.952	the loss after recoveries falls
A fixed budget (1,000 calls)	rank, take the top k, report precision at k	the cost line is stricter than the budget
A stakeholder asks for one number	the leading metric, with the count and the bill beside it	never accuracy alone on an imbalanced outcome